

International Islamic University Chittagong
Center for General Education (CGED)
Midterm Examination, Autumn-2025

Course Title: Basic Principles of Islam
Full Marks: 30

Course Code: GEED-1201
Time: 1.50 Hours

Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question.

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1.	a. Explain briefly (any four): • Meaning of <i>'Aqidah</i> • Articles of Islamic <i>'Aqidah</i> • Meaning of Islam • Kinds of Shirk • Some functions of Angles. Or, b. "Say He is Allah, who is one"- explain this <i>Ayah</i> by focusing on the definition, types, and importance of <i>Tawheed</i>.	10	2	Create Remember & Create
2.	Distinguish between the holy Qur'an and other revealed books with <i>Naqli</i> (Qur'an & Hadith) and <i>'Aqli</i> (logical arguments) evidences.	10	2	Analyze
3.	What are the different stages of Akhirah? Explain the impact of belief in Akhirah on the individual and collective life of Muslims.	10	2	Remember & Create



International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Mid Term Examination, Autumn 2025
Course Code: CSE-1221 Title: Computer Programming 2

Total Marks: 30 Time: 90 minutes

Answer all 3 from the following Questions. Figures in the right-hand margin indicate full marks

Answers must be done sequentially

- | # | | Marks | COs | DL |
|---|--|-------|-----|----|
| 1 | <p>a) Define Object-Oriented Programming (OOP). Describe the four major characteristics of OOP with short examples for each.</p> <p>b) Explain how encapsulation and abstraction help to make programs more secure and modular. Write a small C++ code fragment that demonstrates both concepts in a single class.</p> <p>Or, Explain the difference between structured programming and object-oriented programming in terms of design approach, data handling, and reusability. Write one short C++ code fragment that illustrates how a class encapsulates data and functions together.</p> <p>c) Write a C++ program to create a class Account with private data members <code>acc_no</code>, <code>holder_name</code>, and <code>balance</code>. Use a parameterized constructor to set their values and a member function <code>deposit(double amt)</code> to add money to the balance. In the <code>main()</code> function, make two Account objects, deposit some money into one account, and display both accounts' details. Explain that data hiding is used here because the account details are kept private and can only be changed through class functions, not directly from outside the class.</p> | 3 | CO1 | C1 |
| 2 | <p>a) What is automatic inline function? When should we use inline and when shouldn't? Give an example.</p> <p>b) Write the output of the following C++ program:</p> <pre>#include <iostream> using namespace std; class Sample { public: Sample() { cout << "Default Constructor\n"; } Sample(int x) { cout << "Parameterized Constructor: " << x << endl; } ~Sample() { cout << "Destructor Called\n"; } }; int main() { Sample a, b(10); { Sample c(20); } cout << "End of main\n"; return 0; }</pre> <p>c) Find the error of this code and write correct code.</p> <pre>class Demo { int x; public: Demo(int x) { x = x; } void show() const { cout << x; } ~Demo(int v) { cout << "End"; } void bump(Demo& d); };</pre> | 2 | CO1 | C2 |
| | | 2 | CO2 | C3 |
| | | 4 | CO1 | C2 |
| | | 2 | CO1 | C2 |
| | | 2 | CO2 | C3 |
| | | 2 | CO1 | C2 |
| | | 2 | CO1 | C2 |

```
void bump(Demo d) { d.x++; }
int main() {
    Demo* p = new Demo(5);
    bump(*p);
    cout << p->show() << endl;
    return 0; }
```

- d) Write a C++ program to create a class **Item** with private data members `item_id`, `name`, and `price`. Use a **parameterized constructor** with an **initializer list** to set the values. Add a **static data member** `count` and a **static function** `getCount()` to show how many items have been created. Also include a **friend function** `ApplyDiscount(Item& it, double pct)` to reduce the price by a given percentage. In the `main()` function, create an array of `Item` objects using `new`, display the total count of items, apply a discount to one object using the friend function, print that object's details, and free the memory using `delete[]`. 4 CO2 C3
- Or, Write a C++ program to demonstrate the use of a **constructor**, **destructor**, and **friend function**. Create a class **Rectangle** with private data members `length` and `width`, and add a **parameterized constructor** to initialize these values. Include a **member function** `area()` that calculates and returns the area of the rectangle. Define a **friend function** `compareArea(Rectangle r1, Rectangle r2)` that can access the private members of both objects, compare their areas, and display which rectangle is larger. In the `main()` function, create two `Rectangle` objects with different dimensions and call the `compareArea()` function to show the result.
- 3 a) Define **function overloading**. Explain why overloading improves program readability and flexibility. Describe how the compiler distinguishes between overloaded functions during compilation. Write a short example showing three overloaded functions named **add()** that can add two integers, two doubles, or two strings. 4 CO2 C2
- b) Write a C++ program to show **constructor overloading** using a class **Circle**. Include a **default constructor** that sets the radius to 1.0, a **parameterized constructor** that takes a radius value, and a **copy constructor** that copies the radius from another object. Each constructor should print which one was called. In `main()`, create three `Circle` objects using each constructor and display their radius. 3 CO2 C3
- Or, Write a C++ program to demonstrate a **copy constructor** using a class **Book** with private members `title` and `price`. Use a **parameterized constructor** to set values and a **copy constructor** to create another object with the same data. In `main()`, create one object normally and another using the copy constructor, then display both. Explain briefly when a copy constructor is called automatically.
- c) Analyze the following program and **predict the exact output**. Also, explain briefly how **default arguments** affect function calling. 3 CO1 C1
- ```
void show(int a, int b = 5) {
 cout << "int version: a = " << a << ", b = " << b << endl; }
void show(double x, double y = 2.5) {
 cout << "double version: x = " << x << ", y = " << y << endl; }
int main() {
 show(3);
 show(3, 8);
 show(4.5);
 show(4.5, 6.5);
 return 0; }
```



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**Department of Computer Science and Engineering (CSE)**  
**Autumn- 2025 Mid Term Total Marks-30**

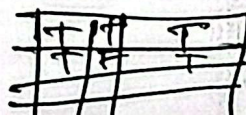
**Course Code: CSE 1223 Course Title: Discrete Mathematics**

**Answer all the following questions. Figures in the right margin indicate full marks.**

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|
| 1. | <p>a) Classify the following as finite and infinite sets. Justify your answer.</p> <p>i) <math>A = \{x : x \in \mathbb{N} \text{ and } x \text{ is even}\}</math></p> <p>ii) <math>C = \{x : x \in \mathbb{N} \text{ and } 3x - 2 = 0\}</math></p> <p>(Note: <math>\mathbb{N}</math> represents the set of Natural Numbers, <math>\{1, 2, 3, \dots\}</math>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 | CLO1 |
|    | <p>b) In a smart home system, devices are organized based on the rooms they are installed in.<br/> <b>Universal Set (all devices in the home):</b><br/> <math>U = \{\text{Smart Light, Smart Thermostat, Smart Lock, Smart Camera}\}</math><br/> <b>Sets:</b></p> <ul style="list-style-type: none"> <li>Set A = {Smart Light, Smart Thermostat} (devices in the living room)</li> <li>Set B = {Smart Thermostat, Smart Lock, Smart Camera} (devices in the entrance)</li> </ul> <p>Represent the following sets as bit strings based on the universal set:<br/> i) Set A    ii) Set B    iii) Set A <math>\oplus</math> B    iv) Set A <math>\cap</math> B</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3 | CLO2 |
| C) | <p>d) Let Set A = {2, 3}, Set B = {a, b}. Find the Cartesian product of Set A and Set B, denoted as A <math>\times</math> B. What is the cardinality of Set A?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1 | CLO2 |
|    | <p>d) In a smart home workshop of 41 participants, it was found that: 23 participants had experience with <b>Smart Lights</b>, 19 participants had experience with <b>Smart Thermostats</b>, 21 participants had experience with <b>Smart Security Cameras</b>, 10 participants had experience with <b>both Smart Lights and Security Cameras</b>, 12 participants had experience with <b>both Smart Lights and Thermostats</b>, 8 participants had experience with <b>both Thermostats and Security Cameras</b>, 6 participants had experience with <b>all three devices</b>. Find the number of participants who had:</p> <p>i) Only Smart Thermostats<br/> ii) Smart Thermostats and Security Cameras but not Smart Lights<br/> iii) Only one of the three smart home devices<br/> iv) None of the three smart home devices<br/> v) At least one of the three smart home devices</p> <p>Or,</p> <p><math>U = \{x \mid x \in \mathbb{Z}^+, 1 \leq x \leq 12\}</math><br/> <math>A = \{x \in U \mid x \text{ is even}\}</math><br/> <math>B = \{x \in U \mid x \text{ is a multiple of } 3\}</math><br/> <math>C = \{x \in U \mid x \text{ is prime and } x &lt; 10\}</math><br/> <math>D = \{x \in U \mid 1 \leq x \leq 4\}</math></p> <p><b>Find the result of the following set operations. List the elements of each resulting set.</b></p> | 5 | CLO1 |



|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |       |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|
|    | i) $((A \cap B) \times (C \cap D))$ ii) $(B \cap D)^c$ iii) $(U - (A \cap B)) \cap (C \cup D)$<br>iv) $A^c \cap D$ v) $(A - B) \cup (B - A)$ vi) $(C \cup D)^c \cap B$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |       |
| 2. | a) <b>Domain:</b> All AI models in a lab<br><b>Predicates :</b> <ul style="list-style-type: none"> <li><math>P(x, y)</math>: "AI model x performs better than AI model y."</li> <li><math>T(x)</math>: "AI model x is trained on a large dataset."</li> </ul> <b>Translate the following logical statements into English:</b> <ul style="list-style-type: none"> <li>i) <math>\exists x \forall y (T(x) \rightarrow P(x, y))</math></li> <li>ii) <math>\forall x \forall y \exists z (P(z, x) \wedge P(z, y))</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | 2 | CLO1  |
|    | b) Consider the following propositions:<br>$x$ : "The AI system is trained on a large dataset."<br>$y$ : "The model achieves high prediction accuracy."<br>$z$ : "The system is deployed for real-world applications."<br>Translate the following propositions into English statements based on the given propositions:<br>i) $x \wedge z$ (Conjunction), ii) $\neg z$ (Negation), iii) $y \leftrightarrow x$ (Biconditional) and iv) $y \rightarrow z$ (Implication)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3 | CLO2  |
|    | c) Given the proposition $((X \rightarrow Y) \wedge (\neg X \rightarrow Y))$ , determine whether the statement is a tautology or not using a truth table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2 | CLO1  |
|    | d) Consider the following predicates: <ul style="list-style-type: none"> <li><math>T(x)</math>: "AI model x is trained on a large dataset."</li> <li><math>A(x)</math>: "AI model x achieves high accuracy."</li> <li><math>D(x, y)</math>: "AI model x is deployed in application y."</li> </ul> Given the domain of all AI models and applications, translate the following English statements into quantifier expressions using the provided predicates: <ul style="list-style-type: none"> <li>i) "All AI models that are trained on large datasets achieve high accuracy."</li> <li>ii) "No AI models that achieve high accuracy are not trained on large datasets."</li> <li>iii) "There exists an AI model x such that if x is trained on a large dataset, then every model y achieves high accuracy; and if x is deployed in some application a, then every model y is deployed in some application b."</li> </ul> | 3 | CLO2  |
| 3  | a) Let p be the function from the set $\{1, 2, 3\}$ to itself such that: $p(1) = 2, p(2) = 3, p(3) = 1$<br>Let q be the function from the set $\{1, 2, 3\}$ to the set $\{10, 20, 30\}$ such that: $q(1) = 30, q(2) = 20, q(3) = 10$ . What is the composition $q \circ p$ ? Is $p \circ q$ defined?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 | CLO2  |
|    | b) Let f and g be functions from $\{1, 2, 3, 4\}$ to $\{a, b, c, d\}$ and from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ , respectively, such that $f(1) = d, f(2) = c, f(3) = a, f(4) = b$ and $g(a) = 2, g(b) = 1, g(c) = 3, g(d) = 2$ .<br>i) Is f one-to-one? Is g one-to-one?<br>ii) Is f onto? Is g onto?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 | CLO 1 |





9) Let the set  $S = \{L, S, C\}$  represent smart home devices. Each relation below is defined on set  $S$ . For each relation, determine whether it is: i) Reflexive ii) Symmetric iii) Antisymmetric iv) Asymmetric v) Transitive

$R1 = \{(L,L), (L,S), (S,L), (S,S), (C,C), (S,C)\}$

$R2 = \{(L,S), (S,C), (C,L)\}$

$R3 = \{(L,L), (S,S), (C,C), (L,S), (S,C), (L,C)\}$

$R4 = \{(L,L), (S,S)\}$

$R5 = \{(L,S), (S,L), (S,C), (C,S)\}$

Or, Let the set  $N = \{11, 12, 13\}$  represent numbers. Each relation below is defined on the set  $N$ . For each relation, determine whether it is: i) Reflexive, ii) Symmetric, iii) Antisymmetric, iv) Asymmetric, v) Transitive

a)  $R1 = \{(12, 13), (12, 12)\}$

b)  $R2 = \{(11, 12), (12, 13), (13, 11)\}$

c)  $R3 = \{(11, 11), (12, 12), (13, 13), (11, 12), (12, 13), (11, 13)\}$

d)  $R4 = \{(11, 11), (11, 12), (12, 11), (12, 12), (13, 13)\}$

4

CLO2



# International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

## Mid Term Examination

Program: B. Sc. in CSE

Course Code: MATH-1207

Time: 1:30 hours



Semester: Autumn 2025

Course Title: Mathematics-II

Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
- (ii) Please answer the several parts of a question sequentially.
- (iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

### Course Learning Outcomes (CLOs) of the Questions

|       |                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| CLO1: | Demonstrate knowledge of geometry and its applications in the real life contexts as well as into complex engineering problems. |
|-------|--------------------------------------------------------------------------------------------------------------------------------|

### Bloom's Taxonomy Domain Levels of the Questions

| Letter Symbols | R        | U          | Ap    | An      | E        | C      |
|----------------|----------|------------|-------|---------|----------|--------|
| Meaning        | Remember | Understand | Apply | Analyze | Evaluate | Create |

- |       |                                                                                                                                                                                                                       | Marks | CLO  | DL |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|----|
| 1. a) | Transform the axes inclined at $30^\circ$ to the original axes the equation,<br>$x^2 + 2\sqrt{3}xy - y^2 = 2a^2$                                                                                                      | 5     | CLO1 | U  |
| Or)   | Find the condition that the lines represented by the homogeneous second degree equation, $ax^2 + 2xy + by^2 = 0$ will be perpendicular with each other.<br>$CD 2axy$                                                  |       |      |    |
| b)    | If the two straight lines represented by $x^2(\tan^2\theta + \cos^2\theta) - 2xy \tan\theta + y^2 \sin^2\theta = 0$ makes angles $\alpha$ and $\beta$ with the axis of x, then show that $\tan\alpha - \tan\beta = 2$ | 5     | CLO1 | U  |
| Or)   | Test the nature of the conic given by the equation,<br>$5x^2 - 24xy - 5y^2 + 4x + 58y - 59 = 0$                                                                                                                       |       |      |    |
| 2. a) | Prove that the angle between the two diagonals of a cube is $\cos^{-1}\left(\frac{1}{3}\right)$                                                                                                                       | 5     | CLO1 | U  |
| b)    | Find the volume of the tetrahedron formed by the planes whose equations are $my + nz = 0$ , $nz + lx = 0$ , $xl + my = 0$ , $lx + my + nz = p$                                                                        | 5     | CLO1 | U  |
| 3. a) | Find the shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$                                                                    | 5     | CLO1 | U  |
| b)    | Find the equation of the sphere for which the circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0$ , $2x + 3y + 4z = 8$ is a great circle.                                                                                      | 5     | CLO1 | U  |

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Department of Computer Science & Engineering  
B. Sc. in CSE Mid Term Examination, Autumn-2025  
Course Code: PHY-1201 Course Title: Physics-II  
Total marks: 30 Time: 1 hours 30 minutes

[Answer *all* the questions. Figures in the right hand margin indicate full marks.]

- |    |    |                                                                                                                                                                                                                                          |      |   |   |
|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|---|
| 1. | a) | State and explain Coulomb's law for electrostatics.                                                                                                                                                                                      | CLO1 | R | 2 |
|    | b) | Derive an expression for electric field due to a long uniformly charged wire.                                                                                                                                                            | CLO2 | U | 5 |
|    | c) | Calculate the electric potential due to a short dipole of dipole moment $4.5 \times 10^{-25}$ coul-metre at a distance of $2 \times 10^{-8}$ cm on a line making an angle $45^\circ$ with the dipole axis from the centre of the dipole. | CLO3 | E | 3 |

- |    |    |                                                                                                                     |      |   |   |
|----|----|---------------------------------------------------------------------------------------------------------------------|------|---|---|
| 2. | a) | State the Faraday's laws of electromagnetic induction.                                                              | CLO1 | R | 2 |
|    | b) | Applying Biot-Savart Law, derive an expression for the magnetic field due to a long straight wire carrying current. | CLO2 | U | 5 |

OR

Define coefficient of self-induction of a coil. Deduce a mathematical expression for the self inductance of a solenoid.

- |    |                                                                                                                                                                                                        |      |   |   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|---|
| c) | A solenoid having a core of cross-section $4 \text{ cm}^2$ , half air and half iron (relative permeability 500), is 22 cm long. If the number of turns on its 1000, what will be its self- inductance? | CLO3 | E | 3 |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|---|

- |    |    |                                                                                                             |      |   |   |
|----|----|-------------------------------------------------------------------------------------------------------------|------|---|---|
| 3. | a) | What is RC Circuit?                                                                                         | CLO1 | R | 1 |
|    | b) | Find out an expression for the growth of charge in capacitor and the decay of current in the circuit.       | CLO2 | U | 6 |
|    | c) | A $30 \mu\text{F}$ capacitor initially uncharged is connected through a 200 ohm resistor to a 12 V battery. | CLO3 | E | 3 |
|    |    | (a) What is the magnitude of the final charge $q_0$ on the capacitor?                                       |      |   |   |
|    |    | (b) What is time constant of the circuit?                                                                   |      |   |   |
|    |    | (c) How long after the capacitor is connected to the battery will it be charged to $\frac{1}{2} q_0$ ?      |      |   |   |

OR

A positive charge of  $q_1 = 2.0 \times 10^{-7} \text{ C}$  is placed at a distance of 0.15 m from another positive charge of  $q_2 = 8.0 \times 10^{-7} \text{ C}$ . At what point on the line joining them is the electric field zero?



**International Islamic University Chittagong**  
**Department of Computer Science and Engineering**

**Mid Term Examination, Autumn-2025**

Course Code: **EEE-1221**

Time: 1 hour 30 minutes

**Program: B.Sc. Engineering in CSE**

Course Title: **Electronics**

Full Marks: 30

[Answer all the questions from the followings. Figures in the right margin indicate full marks]

Course Outcomes (COs) and Blooms Levels are mentioned in additional columns.

- |                                                                                                                                                        | COs | BLOOM'S LEVEL | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|-------|
| 1) a) Explain the working principle of a PN junction diode with proper diagram of its biasing.                                                         | CO1 | U             | 4     |
| 1) b) In the center-tap circuit shown in Fig.1b, the diodes are assumed to be ideal. Find i) d.c output voltage ii) PIV iii) rectification efficiency. | CO2 | Ap            | 6     |

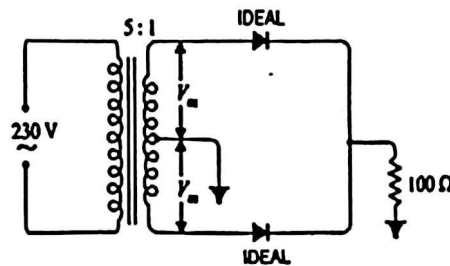


Fig.1b

OR

- |                                                                                                                                                                           |     |    |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|---|
| 1) b) A half-wave rectifier is used to supply 45V d.c. to a resistive load of 800 $\Omega$ . The diode has a resistance of 15 $\Omega$ . Calculate a.c. voltage required. | CO2 | Ap | 6 |
| 2) a) Draw the symbol of an N-P-N transistor and briefly explain its construction and working principle.                                                                  | CO1 | U  | 4 |
| 2) b) Calculate the emitter current in the voltage divider circuit shown in Fig.2b. Also find the value of $V_{CE}$ and collector potential $V_C$ .                       |     |    |   |

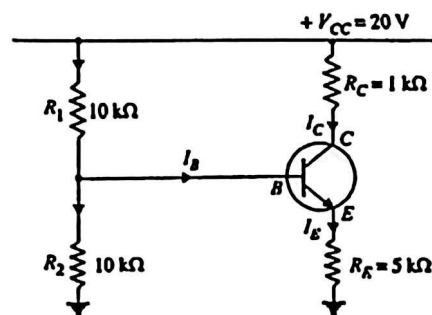


Fig.2b

- |                                                                                                                            |     |   |   |
|----------------------------------------------------------------------------------------------------------------------------|-----|---|---|
| 3) a) What is linear and non linear wave shaping ? Design and explain the output waveform for a Parallel negative clipper. | CO1 | E | 5 |
| OR                                                                                                                         |     |   |   |
| 3) a) What is biased diode clipping circuit ? Design and explain the output waveform for a negative biased diode clipping. | CO1 | E | 5 |

- 3) b) Explain the working of a clamper circuit. A clamper circuit is shown in Fig. 3(b). Find the output waveform for the input wave  $V_{in}$ .

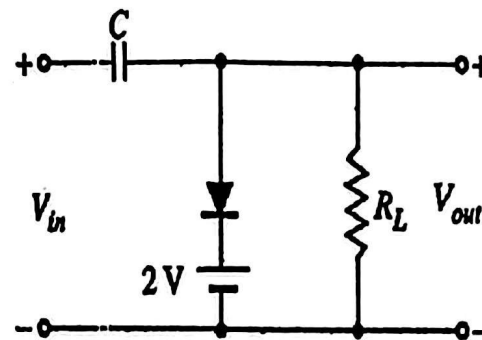
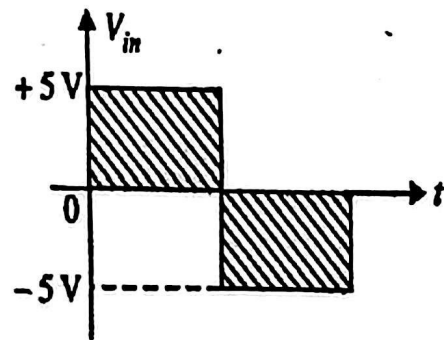
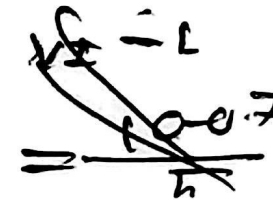


Fig.3b



CO2 An 5